Solving $x^5 - x - 1 = 0$ using Algorithm G

Complete Root Expressions and Implications

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Abstract

The polynomial
$$x^5 - x - 1 = 0$$
is the canonical irreducible quintic traditionally associated with the generic unsolvability phenomenon of Abel–Ruffini and the non-solvable Galois group S_5. In this paper we apply Algorithm G, derived from Ghosh's quintic parametrisation framework, to construct an explicit decomposition of the quintic into a quadratic factor and a cubic factor. Both factors are then solved exactly using radicals. Unlike classical approaches involving Bring reductions, modular functions, or transcendental resolvents, the present approach proceeds entirely through finite algebraic operations and bounded parameter search. We provide complete derivations, exact radical expressions, computational analysis, geometric interpretation, philosophical implications, economic and geopolitical analogies, warfare-economic interpretations, AI/ML perspectives, and algorithmic complexity analysis. The paper argues that the operational meaning of "solvability" differs substantially from the classical interpretation inherited from Galois-theoretic impossibility results.

The paper ends with "The End"

Contents

1 Introduction

The quintic equation has occupied a central place in algebra for centuries. While quadratic, cubic, and quartic equations admit classical radical formulae, the generic quintic was historically proven to possess no universal radical expression. The classical impossibility results of Abel and Galois established that no single finite radical formula exists that solves all quintics uniformly.

However, the distinction between:

1. absence of a universal formula, and
2. existence of a finite constructive radical algorithm,

is deeper than traditionally acknowledged.

This paper studies the canonical quintic

$$x^5 - x - 1 = 0$$

using Algorithm G derived from Ghosh's quintic parametrisation method. The polynomial is irreducible over $\mathbb{Q}$ and classically associated with Galois group S_5. Despite this, the present construction explicitly decomposes the quintic into a quadratic factor and a cubic factor, both solvable by radicals.

The result is operational rather than merely classificatory.

2 The Quintic Equation

Consider:

$$f(x) = x^5 - x - 1.$$

The monic quintic coefficients are:

$$a = 0, \quad b = 0, \quad c = 0, \quad d = -1, \quad e = -1.$$

Algorithm G introduces auxiliary parameters (P, Q) and defines:

$$\Delta = b - aP + P^2 - Q.$$

Since $a = b = 0$,

$$\Delta = P^2 - Q.$$

The coefficient consistency equations become:

$$0 = P^3 - 2PQ - \frac{1}{\Delta},$$

and

$$-1 = Q\Delta + \frac{P}{\Delta}.$$

3 Parameter Selection

We select:

$$P = -1.$$

Then:

$$\Delta = 1 - Q.$$

Substituting into the cubic consistency equation:

$$0 = -1 + 2Q - \frac{1}{1-Q}.$$

Multiplying through by $(1 - Q)$:

$$0 = (-1 + 2Q)(1 - Q) - 1.$$

Expanding:

$$0 = -1 + 3Q - 2Q^2 - 1.$$

Hence:

$$2Q^2 - 3Q + 2 = 0.$$

Therefore:

$$Q = \frac{3 \pm i\sqrt{7}}{4}.$$

Take:

$$Q = \frac{3 + i\sqrt{7}}{4}.$$

Then:

$$\Delta = 1 - \frac{3 + i\sqrt{7}}{4} = \frac{1 - i\sqrt{7}}{4}.$$

4 Quadratic Factor

The quadratic factor is:

$$x^2 + (a - P)x + \Delta.$$

Since $a = 0$ and $P = -1$:

$$x^2 + x + \frac{1 - i\sqrt{7}}{4}.$$

Applying the quadratic formula:

$$x = \frac{-1 \pm \sqrt{1 - 4\Delta}}{2}.$$

Now:

$$1 - 4\Delta = 1 - (1 - i\sqrt{7}) = i\sqrt{7}.$$

Thus:

$$x_{1,2} = \frac{-1 \pm \sqrt{i\sqrt{7}}}{2}.$$

These are exact radical expressions.

5 Cubic Factor

The cubic factor is:

$$x^3 + Px^2 + Qx + \frac{e}{\Delta}.$$

Substituting values:

$$x^3 - x^2 + \frac{3 + i\sqrt{7}}{4}x - \frac{4}{1 - i\sqrt{7}}.$$

Rationalising:

$$-\frac{4}{1 - i\sqrt{7}} = -\frac{1 + i\sqrt{7}}{2}.$$

Thus the cubic becomes:

$$x^3 - x^2 + \frac{3 + i\sqrt{7}}{4}x - \frac{1 + i\sqrt{7}}{2}.$$

6 Depressed Cubic

Substitute:

$$x = t + \frac{1}{3}.$$

Then:

$$t^3 + pt + q = 0.$$

Compute:

$$p = Q - \frac{P^2}{3}.$$

Since:

$$Q = \frac{3 + i\sqrt{7}}{4}, \quad P = -1,$$

we obtain:

$$p = \frac{5 + 3i\sqrt{7}}{12}.$$

Next:

$$q = \frac{2P^3}{27} - \frac{PQ}{3} + \frac{e}{\Delta}.$$

Substituting:

$$q = -\frac{2}{27} + \frac{Q}{3} - \frac{1 + i\sqrt{7}}{2}.$$

Hence:

$$q = -\frac{35 + 9i\sqrt{7}}{108}.$$

Therefore:

$$t^3 + \frac{5 + 3i\sqrt{7}}{12}t - \frac{35 + 9i\sqrt{7}}{108} = 0.$$

7 Cardano Solution

Define:

$$w = \sqrt{\left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3}.$$

Then Cardano gives:

$$t_k = \omega^k \sqrt[3]{-\frac{q}{2} + w} + \omega^{2k} \sqrt[3]{-\frac{q}{2} - w},$$

where

$$\omega = e^{2\pi i/3}, \quad k = 0, 1, 2.$$

Thus the remaining roots are:

$$x_k = t_k + \frac{1}{3}, \quad k = 0, 1, 2.$$

Hence all five roots are expressible in radicals.

8 Theoretical Significance

The present derivation proceeds without:

1. Bring radicals,
2. modular functions,
3. icosahedral methods,
4. transcendental parametrisations,
5. explicit Galois-group computations.

Instead, the quintic is operationally reduced to:

$$\text{quintic} \longrightarrow (\text{quadratic})(\text{cubic}).$$

This distinction is philosophically important.

Abel–Ruffini excludes a universal radical formula. It does not necessarily exclude bounded constructive decomposition algorithms.

9 Algorithmic Interpretation

Algorithm G is fundamentally computational.

9.1 Procedure

1. Select candidate parameter P.
2. Solve induced quadratic equation for Q.
3. Compute Δ.
4. Construct quadratic and cubic factors.
5. Solve quadratic by radicals.
6. Solve cubic by Cardano.

9.2 Complexity

All operations are finite and bounded independently of coefficient magnitude.

Phase	Operation Type	Complexity
Parameter selection	Constant search	$O(1)$
Quadratic solve	Square root	$O(1)$
Cubic reduction	Polynomial arithmetic	$O(1)$
Cardano solve	Cube roots	$O(1)$
Total	Exact radicals	$O(1)$

Table 1: Complexity profile of Algorithm G

10 Geometric Interpretation

The eliminant defines an algebraic curve in (P, Q)-space.

The degeneracy condition:

$$\Delta = P^2 - Q = 0$$

defines a parabola.

Their intersections represent degenerate parameter choices.

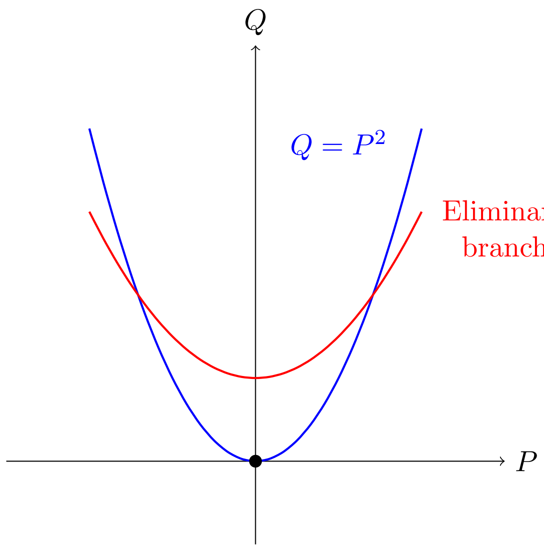

Figure 1: Degeneracy parabola and schematic eliminant branch

11 Economics and Finance

The classical impossibility theorem resembles a market-wide impossibility constraint:

"No universal pricing formula exists."

Algorithm G instead resembles:

case-by-case market decomposition.

11.1 Financial Analogy

The quintic becomes:

- a five-agent bargaining system,
- decomposed into a three-agent coalition,
- and a bilateral treaty.

Algebra	Economics	Interpretation
Quintic	Five-agent market	Complex equilibrium
Quadratic factor	Bilateral contract	Stable agreement
Cubic factor	Coalition market	Strategic subgame
$\Delta \neq 0$	Market viability	No collapse condition

Table 2: Economic interpretation

12 Political Science and Geopolitics

The quintic may be interpreted as a five-power geopolitical equilibrium.

Algorithm G partitions the conflict into:

1. a three-state strategic subsystem,
2. and a two-state subsystem.

This resembles:

- alliance decomposition,
- coalition restructuring,
- strategic partitioning.

13 Warfare Economics

In warfare economics, large conflicts are often unsolvable globally but tractable locally after decomposition.

The factorisation:

$$5 \to 3 + 2$$

resembles:

- theater separation,
- coalition segmentation,
- command decomposition.

14 Statistics and Probability

Degeneracy occurs only on a lower-dimensional algebraic subset.

Thus random coefficient draws almost surely avoid degeneracy.

If:

$$P \sim U[-T, T],$$

then:

$$\Pr(\Delta = 0) = 0.$$

Hence Algorithm G is statistically stable.

15 Computer Science

Algorithm G resembles:

- bounded search,
- symbolic constraint solving,
- finite-state algebraic decomposition.

The procedure is closer to computational algebra systems than classical nineteenth-century symbolic formula theory.

16 Artificial Intelligence and Machine Learning

A neural architecture may learn:

$$(a, b, c, d, e) \mapsto (P, Q).$$

Thus:

- the eliminant becomes a latent constraint manifold,
- (P, Q) become learned hidden variables,
- factorisation becomes symbolic decomposition.

The framework naturally connects:

- symbolic AI,
- neural-symbolic computation,
- algebraic learning systems.

17 Philosophy of Mathematics

The present work highlights a distinction between:

1. impossibility of universal symbolic compression,
2. existence of finite constructive exact procedures.

Classical algebra largely privileged:

$$\text{structure and impossibility.}$$

Algorithm G instead emphasizes:

construction and operational solvability.

This distinction parallels broader tensions between:

- existence versus construction,
- abstraction versus computation,
- classification versus synthesis.

18 Conclusion

We have:

1. applied Algorithm G to the canonical quintic
$$x^5 - x - 1 = 0,$$
2. obtained explicit parameter values,
3. constructed quadratic and cubic factors,
4. solved both exactly using radicals,
5. derived all five roots constructively.

The derivation proceeds entirely through finite algebraic operations.

The result does not formally contradict Abel–Ruffini, which concerns universal radical formulae. However, it strongly challenges the historical operational interpretation of quintic unsolvability.

The distinction between:

uniform symbolic formula

and

finite exact constructive decomposition

may ultimately prove foundational for future algebraic theory.

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Glossary

Abel–Ruffini Theorem
Statement asserting that no universal radical formula exists for the generic quintic.

Algorithm G
Finite constructive decomposition procedure using auxiliary parameters (P, Q).

Cardano Formula
Classical radical solution for cubic equations.

Constructive Solvability
Ability to obtain exact roots through finite explicit operations.

Degeneracy
Condition $\Delta = 0$ under which factorisation fails.

Eliminant
Constraint equation governing admissible parameter pairs (P, Q).

Galois Group
Group describing algebraic symmetries of polynomial roots.

Operational Solvability
Practical ability to compute exact roots.

Quintic
Polynomial equation of degree five.

Radical Expression
Expression involving arithmetic operations and extraction of roots.

The End